

# 24CS31 — Last-Minute Prediction Sheet

Laplace Transforms & Vector Space · SEE 100 marks / 3 hrs · **Answer one full question from each unit** · Q1|Q2 → Unit I Q3|Q4 → II Q5|Q6 → III Q7|Q8 → IV Q9|Q10 → V

**THE 60-SECOND RULE.** Don't read all ten questions. In each unit, find the **one trigger** below and answer **that** question — the rest of its parts are the ones you studied. **I:** the  $t^n$  proof • **II:** the Heaviside piecewise • **III:** the reflection→rotation→contraction question • **IV:** QR + least-squares (that's Q8) • **V:** orthogonally diagonalize. Frequency key: **4/4** in all four papers **3/4** **2/4** **1/4**

## UNIT I — Laplace Transforms

20 marks · a(2) + b(4) + c(7) + d(7)

**PICK the question containing the  $t^n$  proof.** The  $\sin \sqrt{t}$  question is **always** in the same one (4/4 papers) — that's 11 marks together. The periodic-function problem is always in the *other* one.

| # | Question — write this                                                                                                                                                                                                                                   | Marks | Section        | Freq |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------|------|
| 1 | <b>Prove</b> $L\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} \{F(s)\}$ , $n$ a positive integer. ← <b>THE trigger. Do this first.</b>                                                                                                                          | 07    | Q1c or Q2b     | 4/4  |
| 2 | Given $L\{\sin \sqrt{t}\} = \frac{\sqrt{\pi}}{2s^{3/2}} e^{-1/4s}$ , <b>show</b> $L\{\frac{\cos \sqrt{t}}{\sqrt{t}}\} = \frac{\sqrt{\pi}}{s^{1/2}} e^{-1/4s}$                                                                                           | 04    | Q1b — with #1  | 4/4  |
| 3 | <b>Evaluate</b> (i) $L\{t e^{-4t} \sin 3t\}$ (ii) $L\{\frac{t - \sinh at}{t}\}$ (multiply by $t \rightarrow -F(s)$ ; divide by $t \rightarrow \int_s^\infty F$ )                                                                                        | 07    | Q1d or Q2c     | 4/4  |
| 4 | <b>Evaluate</b> (i) $\int_0^\infty \frac{e^{-t} \sin t}{t} dt$ (ans $\pi/4$ ) (ii) $L\{t \sin 3t \cos 2t\}$                                                                                                                                             | 07    | Q1d            | 3/4  |
| 5 | <b>Periodic function.</b> Triangular wave, period $2a$ : $f(t) = \begin{cases} t/2a - t, & 0 \leq t \leq a \\ a \leq t \leq 2a \end{cases}$ ans $\frac{1}{s^2} \tanh(\frac{as}{2})$ OR half-rectifier $E \sin(\omega t)$ on $(0, \pi/\omega)$ , 0 after | 07    | OTHER question | 4/4  |
| 6 | <b>Obtain</b> $L\{(\sqrt{t} + 1/\sqrt{t})^3\}$ (expand first, then term-by-term)                                                                                                                                                                        | 04    | Q2b            | 2/4  |
| 7 | <b>Definition (2 marks — free).</b> “Write the Laplace transform of a periodic function” = $\frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt$ • or “Define the Laplace transform”                                                                         | 02    | Q1a & Q2a      | 4/4  |

## UNIT II — Application of Laplace Transforms

20 marks · a(2) + b(4) + c(7) + d(7)

**PICK the question containing the Heaviside piecewise function.** An ODE problem is **always** sitting next to it (4/4 papers) — that's **14 guaranteed marks** in one question. Convolution is in the other one.

| # | Question — write this                                                                                                                                                                                                                                                                                                     | Marks | Section             | Freq |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------------------|------|
| 1 | <b>Express in terms of the Heaviside function and find its LT:</b> $f(t) = \begin{cases} t^2, & 0 < t < 2 \\ 4t, & 2 < t < 4 \\ 8, & t > 4 \end{cases}$ ← <b>THE trigger. Same values twice.</b>                                                                                                                          | 07    | Q3c                 | 4/4  |
| 2 | <b>Solve by Laplace transforms</b> (one of these, in the <i>same</i> question as #1):<br>• $dx/dt - 2y = \cos 2t$ ; $dy/dt + 2x = \sin 2t$ , $x(0)=1$ , $y(0)=0$<br>• $y'' + 4y' + 3y = e^{-t}$ (set twice, different ICs)<br>• LR circuit: $L di/dt + Ri = Ee^{-at} \rightarrow i = \frac{E}{R-aL}(e^{-at} - e^{-Rt/L})$ | 07    | Q3d / Q4d — with #1 | 4/4  |
| 3 | <b>Convolution theorem.</b> $L^{-1}\{\frac{s^2}{(s^2+a^2)(s^2+b^2)}\}$ OR verify it for $f_1=t$ , $f_2=\cos t$                                                                                                                                                                                                            | 07    | OTHER question      | 4/4  |
| 4 | <b>Evaluate</b> $L^{-1}\{\log \frac{s^2+1}{s(s+1)}\}$ differentiate $F$ , then $L^{-1}\{F\} = -t f(t)$                                                                                                                                                                                                                    | 04    | Q3b                 | 3/4  |
| 5 | <b>Find</b> $L^{-1}\{\frac{3s+2}{s^2-s-2}\}$ — partial fractions. Every paper carries a short inverse LT in <i>both</i> options.                                                                                                                                                                                          | 04    | Q4b                 | 4/4  |
| 6 | <b>Definition (2 marks — free).</b> Define the unit step function + graph • Define the Dirac-delta function + sketch                                                                                                                                                                                                      | 02    | Q3a & Q4a           | 4/4  |

## UNIT III — Vector Space and Linear Transformation

20 marks · a(6) + b(7) + c(7)

**PICK the question containing the reflection→rotation→contraction problem** (usually **Q6**). In the two newest papers that question also held the transition matrix *and* kernel&range — three 4/4 topics, a clean 20.

| # | Question — write this                                                                                                                                                                                                                                                                                                                           | Marks | Section            | Freq |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------------------|------|
| 1 | <b>Matrix of a composition + image of a point.</b> Reflection in the $x$ -axis, then rotation $\pi/2$ , then contraction of factor $1/3$ . Find the image of $\begin{bmatrix} 4 \\ 1 \end{bmatrix}$ . ← <b>THE trigger.</b><br>Multiply <b>right to left</b> : $k R_Q$ Ref. Reflect in $x = \text{diag}(1, -1)$ ; in $y = \text{diag}(-1, 1)$ . | 06    | Q6a                | 4/4  |
| 2 | <b>Transition matrix.</b> $B = \{(1, 2), (3, -1)\}$ , $B' = \{(3, 1), (5, 2)\}$ of $R^2$ . Find the transition matrix from $B$ to $B'$ ; if $[u]_B = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ , find $[u]_{B'}$ . <b>Same bases twice running.</b>                                                                                                 | 07    | Q6b — with #1      | 4/4  |
| 3 | <b>Kernel and range + verify rank-nullity</b> for $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & 1 \\ 1 & 1 & 4 \end{bmatrix}$ (rank 2, nullity 1). Be ready for the formula version too, e.g. $T(x, y) = (3x, x-y, y)$ .                                                                                                                           | 07    | Q6c — with #1, #2  | 4/4  |
| 4 | <b>Prove <math>T</math> is linear + find images.</b> $T: P_2 \rightarrow P_2$ , $T(ax^2 + bx + c) = (a+b)x^2 + c$ ; image of $5x^2 + 6x + 1$ . OR $T(x, y) = (3x+y, 2y, x-y)$ ; images of $(1, 2)$ , $(2, -5)$                                                                                                                                  | 07    | Q5b — all 4 papers | 4/4  |
| 5 | <b>Span / linear combination / independence / basis.</b> e.g. is a set of four $2 \times 2$ matrices a basis of $M_{2,2}$ ? (flatten each to a 4-vector, take the $4 \times 4$ determinant) • or: does $\{(1, 2, 3), (-1, -1, 0), (2, 5, 4)\}$ span $R^3$ ?                                                                                     | 06    | Q5a                | 4/4  |
| 6 | <b>Prove</b> $T(x) = Ax$ rotates $x$ through $\theta$ about the origin if $A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ — bookwork escape hatch if Q6 looks ugly.                                                                                                                                     | 07    | Q5c                | 2/4  |

## UNIT IV — Orthogonal Projections

20 marks · Q7 = 6+7+7 | Q8 = 10+10

**Q8 is QR + least-squares** — both 4/4, and that's the whole 20. Take it if your algebra holds up. **Q7 is the safer three-problem question** — and the four-subspaces question is *always* in Q7 (4/4).

| # | Question — write this                                                                                                                                                                                                                                                                         | Marks | Section        | Freq |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------|------|
| 1 | <b>QR factorization.</b> Mar 24 used $\begin{bmatrix} 3 & -5 & 1 \\ 1 & 1 & 1 \\ -1 & 5 & -2 \\ 3 & -7 & 8 \end{bmatrix}$ . Matrix changes every paper — own the method: Gram-Schmidt the columns, normalise $\rightarrow Q$ , then $R = Q^T A$ .                                             | 10    | Q8a            | 4/4  |
| 2 | <b>Least-squares solution of <math>AX = b</math>:</b> $A = \begin{bmatrix} 1 & 3 & 5 \\ 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 3 & 3 \end{bmatrix}$ $b = \begin{bmatrix} 3 \\ 5 \\ 7 \\ -3 \end{bmatrix}$ ← <b>identical values, twice.</b><br>Solve $A^T A x = A^T b$ .                               | 10    | Q8b — with #1  | 4/4  |
| 3 | <b>Dimension and basis for the four fundamental subspaces</b> of $A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix}$ ← <b>same matrix in 3 of 4 papers.</b> row 3 = row 1 $\Rightarrow$ rank 2; dim nul = 2, dim left-nul = 1                                 | 07    | Q7 — always    | 4/4  |
| 4 | <b>Complete solution of <math>Ax = b</math>.</b> Mar 24: $x+3y+3z=1$ , $2x+6y+9z=5$ , $-x-3y+3z=5$ . Answer as $x = x_p + c_1 n_1 + \dots$                                                                                                                                                    | 06    | Q7a            | 3/4  |
| 5 | <b>Orthogonal projection.</b> $y = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ , $u = \begin{bmatrix} 4 \\ -7 \end{bmatrix}$ . Find proj of $y$ onto $u$ ; split $y$ into a part in $\text{span}\{u\}$ and a part orthogonal to it. <b>proj</b> = $((y \cdot u)/(u \cdot u))u$ — 5 minutes, 7 marks | 07    | Q7b            | 1/4  |
| 6 | <b>Gram-Schmidt orthonormal basis</b> — e.g. $\text{span}\{(1, 1, 1, 1), (0, 1, 1, 1), (0, 0, 1, 1)\}$ . You already know this from QR: just stop before forming $R$ .                                                                                                                        | 10    | Q8a substitute | 1/4  |

## UNIT V — Applications of Eigenvalue Decomposition

20 marks · a(5) + b(5) + c(10)

**PICK the question containing “orthogonally diagonalize”** (Q10 in Mar 24). The two property-checks travel with it in 3 of 4 papers — 10 + 5 + 5, and the 5-markers are ten minutes total. SVD sits in the *other* question.

| # | Question — write this                                                                                                                                                                                                                                                                                             | Marks | Section            | Freq |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------------------|------|
| 1 | <b>Orthogonally diagonalize</b> $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$ ← <b>THE trigger. Same matrix, 3 of 4 papers.</b><br>eigenvalues <b>2, 3, 6</b> — all distinct, so eigenvectors come out orthogonal on their own.<br>Normalise $\rightarrow P$ , then $P^T A P = D$ . | 10    | Q10c               | 4/4  |
| 2 | <b>Is</b> $A = \begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix}$ <b>positive definite?</b> minors 1, 0, 0 $\rightarrow$ <b>positive semi-definite, NOT positive definite.</b> Say so explicitly.                                                                                             | 05    | Q10a — with #1     | 4/4  |
| 3 | <b>Is</b> $A = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}$ <b>unitary?</b> Show $A^* A = I$ (yes) OR the Hermitian version: is $\begin{bmatrix} 3 & 7-4i & -2+5i \\ 7+4i & -2 & 3+i \\ -2-5i & 3-i & 4 \end{bmatrix}$ Hermitian? (yes)                                                                         | 05    | Q10b — with #1, #2 | 4/4  |

|   |                                                                                                                                                                                                                |    |                      |     |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----------------------|-----|
| 4 | Find the SVD of $A = \begin{bmatrix} 1 & 1 \\ 3 & -3 \end{bmatrix}$ $A^T A = \begin{bmatrix} 10 & -8 \\ -8 & 10 \end{bmatrix}$ , eigenvalues 18 and 2 $\Rightarrow$ singular values $3\sqrt{2}$ and $\sqrt{2}$ | 10 | OTHER question (Q9c) | 4/4 |
| 5 | Quadratic form $\rightarrow$ no cross-product terms. $Q(x) = x_1^2 - 8x_1x_2 - 5x_2^2$ . Build the symmetric matrix, orthogonally diagonalise, substitute $x = Py$ .                                           | 05 | Q9a                  | 2/4 |
| 6 | Steady state of the Markov matrix $A = \begin{bmatrix} 0.95 & 0.03 \\ 0.05 & 0.97 \end{bmatrix}$ solve $(A-I)x = 0$ , scale so entries sum to 1                                                                | 05 | Q9b                  | 1/4 |
| 7 | Diagonalize and hence find $A^n$ ( $A^n = PD^nP^{-1}$ ). Appeared 3 papers straight, then skipped in Mar 24 — which makes it <b>overdue</b> , not safe.                                                        | 10 | Q9 (10-mark slot)    | 3/4 |

**IF YOU HAVE 20 SPARE MINUTES — the one hole in your prep.** PCA is named in your syllabus *and* in CO5, and has **never** been asked in four papers. If it comes, it's 5 marks in Unit V, most likely next to the SVD. All you need: centre the data (subtract the column means), form the covariance matrix  $C = \frac{1}{n-1} X^T X$ , find its eigenvalues and eigenvectors, take the eigenvector of the largest eigenvalue as the first principal component, and report  $\lambda_1/\Sigma\lambda_i$  as the variance it explains. **Conic sections** is the same gap — diagonalise the quadratic form; both eigenvalues the same sign  $\rightarrow$  ellipse, opposite signs  $\rightarrow$  hyperbola.

**How to read this:** “Section” is the slot the question occupied in the March 2024 paper (the only one under your current course title) with the older papers cross-checked. “Freq” is how many of the four papers carried that topic. Items marked **identical values** have been set with the same numbers more than once. Predictions are inferences from four papers — strong, not guaranteed. Full 22-page analysis, pairing laws and the complete question index are in *24CS31\_Prediction\_Sheet.pdf*.